Counterexample to conjecture 4.12 of Akbari, Elphick, Kumar, Pragada, and Tang

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1 Conjecture

Conjecture 1 (conjecture 4.12 [1])**.** *For every connected graph G with line graph $L(G)$,*

$$n^+(L(G)) \leq n^-(L(G)) + 1, \qquad \textit{that is} \quad s(L(G)) \leq 1.$$

The authors verify the conjecture on numerous graphs such that $m(G) = n(L(G)) \leq 100$, and find no counterexample among graphs of order at most 9. They [1] state that the conjecture holds whenever $m \geq 2n - 1$. However, it remains open when $n \leq m \leq 2n - 2$.

2 Results

2.1 Graph identification

The graph is defined by the following graph6 string:

```
‘?G??O__???G?DOCAO?_???G?_C?AA?G?_GCA?_?C??G??????R???_?‘?A???o?@A??????_?????????A????@A
```

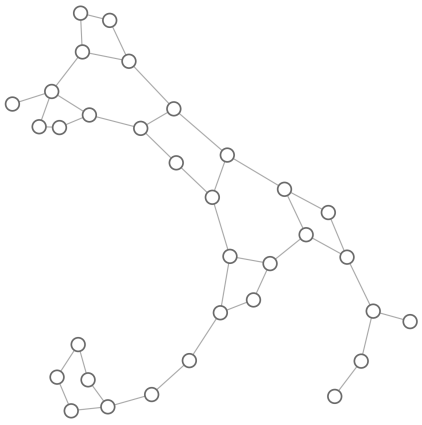


Figure 1: Counterexample

2.2 Calculated properties

Property	Value
Number of vertices $n(G)$	33
Number of edges $m(G) = n(L(G))$	40
$n^+(L(G))$	21
$n^0(L(G))$	0
$n^-(L(G))$	19
Signature $s(L(G))$	2
Conjectured bound $n^-(L(G)) + 1$	20

Table 1: Properties of G and its line graph $L(G)$.

3 Verification of the inequality

We first verify that G satisfies the condition of the conjecture. With $n(G) = 33$ and $m(G) = 40$, we have:

$$n(G) \leq m(G) \leq 2\,n(G) - 2, \qquad \text{that is,} \qquad 33 \leq 40 \leq 64.$$

Computation of the left-hand side (LHS):

$$\text{LHS} = n^+(L(G)) = 21.$$

Computation of the right-hand side (RHS):

$$\text{RHS} = n^-(L(G)) + 1 = 19 + 1 = 20.$$

Verification:

$$21 \leq 20 \quad \Longrightarrow \quad \textbf{FALSE}$$

4 Conclusion

The conjecture states that $n^+(L(G)) \leq n^-(L(G)) + 1$. However, we observe:

$$\text{LHS} = 21 \quad \text{and} \quad \text{RHS} = 20,$$

$$21 > 20$$

The inequality is not satisfied. **This graph is therefore a valid counterexample.**

References

[1] S. Akbari, C. Elphick, H. Kumar, S. Pragada, and Q. Tang, *A new conjecture on the inertia of graphs*, arXiv:2508.01163 (2025).